Assignment Title	Group Assignment
Learning outcomes covered	LO 2 - 4
Assignment Mode	Project
Maximum Marks	100 marks
Contribution to the Final Grade	20%
Date published	3rd Week
Deadline for submissions	3rd Week, 10th Week, 13th Week
Mode of Submission	Group Presentation and Report

Assignment Description

Overview

This assignment is designed to help you apply software engineering principles, including SDLC stages, Agile practices, UML modeling, and design patterns, in a practical team-based project. You will work in pre-assigned groups to design and develop a real-world system using Java. Over the course of the semester, your team will go through multiple development phases, create a functional prototype, apply Agile (Scrum), demonstrate your progress, and submit a final working system along with the required documentation.

- Groups: Already formed (6 members each)
- Platform: Java-based web application
- Assignment Weight: 20% of final module grade

Note: In this assignment, students will be assessed both individually and as part of a team across all evaluation components.

Deliverables & Timeline

Phase	Week	Deliverables	Marks	Weightage
Proposal Evaluation	Week 3	Proposal Presentation	70 marks	20%
		Report	30 marks	
Progress Evaluation	Week 10	Progress Presentation	50 marks	30%
		Design Document	50 marks	
Final Evaluation and Viva	Week 13	Final Presentation and Viva	70 marks	50%
		Final Report	30 marks	

Project Life Cycle and Deliverables

Phase 1: Project Proposal

- Week: 3
- Weight: 20% of the total project mark
- Components: Presentation and Proposal Report

Proposal Presentation Instructions

- Duration: 10 minutes presentation and 5 minutes Q&A
- Participation: All group members must actively participate. Each member must present at least their assigned major function.

Proposal Presentation Content

Your presentation should clearly and concisely cover the following:

Presentation Section	Details
1. Project Title and Group Members' Names	Mention the system name and list all group members.
2. Introduction	Brief overview of the project - Objectives and purpose
3. System Overview Diagram	A visual showing the overall system flow
4. Functional Requirements	Key features and functionalities of the system
5. Non-Functional Requirements	Qualities like performance, usability, security, etc.
6. Major Stakeholders / Users	Who will use or benefit from the system
7. Six Major Functions (one per member)	Identify the user of each function - Explain the main tasks involved in each function
8. Minor Functions	Supporting features such as login, logout, etc.
9. System Limitations / Constraints	Any known assumptions, technical boundaries, or restrictions
10. Project Timeline	Weekly plan from Week 3 to Week 14 (include key expected activities)
11. Conclusion	Final remarks on the importance, feasibility, and value of the proposed system

Proposal Report Instructions

- **Submission Format**: Submit the report as a **soft copy (PDF format)** through the designated platform.
- **Length**: Maximum 12–15 pages
- **Submission**: One report per group. Submit soft copy through the course platform
- **Purpose**: This report is a formal document that supports your proposal presentation. It should clearly outline your system idea, user needs, and project plan.

Proposal Report Structure

Section (with Suggested Page Count)	Details
1. Cover Page (1 page)	Include: • Project Title • Group Members' Names and Student IDs • Module Name and Code (e.g., SE2030 – Software Engineering) • Submission Date
2. Introduction (1–2 pages)	Provide a clear overview of the project. Include: • What is the project about? • Objectives or goals • Target users or stakeholders • Scope and limitations
3. System Overview Diagram (0.5–1 page)	Add a high-level diagram showing how your system will work. It can be a simple block or flow diagram.
4. Functional Requirements (1 page)	Clearly list the main functions your system must support. (e.g., user registration, booking, notifications)
5. Non-Functional Requirements (0.5–1 page)	Define quality aspects such as performance, security, usability, or scalability.
6. Major Stakeholders / Users (0.5–1 page)	Identify user roles like admin, customer, staff, suppliers, etc.
7. Six Major Functions (2 pages)	Assign one function to each group member. For each, describe:

	• Who will use it • What it does • Expected outcome
8. Minor Functions (0.5 - 1 page)	Mention common supportive functions (e.g., login, password reset).
9. System Limitations / Constraints (0.5 – 1 page)	Mention known constraints such as platform limitations, expected device type, or assumptions.
10. Simple Timeline (0.5–1 page)	Create a basic timeline table from Week 3 to Week 14. For each key week, include what your group *plans to complete* during that time.
11. Conclusion (0.5–1 page)	Summarize your project idea and emphasize why it's worth implementing. Mention how your team plans to successfully complete it.

Phase 2: Progress Presentation and Design Document

- Week: 10
- Weight: 30% of the total project mark (30 Marks)
- Components: System demonstration and Design Document
- Duration: 15 minutes System Demonstration and 5 minutes Q&A
- 75% of the project should be complete at this level.
- Participation: All group members must contribute and explain their progress, especially on their assigned major function.
- Refer to the marking rubric to understand what will be evaluated during the progress presentation, under both the system demonstration and the design document sections.

Design Document Preparation and Submission Instructions

- **Submission Format**: Submit the document as a **soft copy (PDF format)** through the course web links.
- **Length**: Maximum 12–15 pages
- **Submission**: One report per group. Submit soft copy through the course platform
- **File name:** <<Group ID>>.pdf

Design Document Structure

Component	Details
1. Agile Sprint Summaries	• Provide a summary of sprints completed so far • Provide summaries of sprints to be completed (Mention key deliverables and activities per sprint)
2. Final Use Case Diagram	• Reflect Users and corresponding functions
3. Class Diagram	• Show object-oriented structure and key relationships
4. Activity Diagrams	• One Activity Diagram for each major function
5. Ethical Considerations	• Describe any ethical concerns and how they are addressed • Example: data privacy, user consent, accessibility

Phase 3 - Final Presentation

- Week: 13
- Duration: 20 minutes System demonstration and 10 minutes Q&A
- Type: Team-based presentation and demonstration
- Expectations:
    - Demonstrate a fully working system
    - Each team member should present part of the system, especially their assigned major function
    - Explain how the system meets the defined requirements and objectives
    - Highlight key features, technologies used, and any improvements made based on earlier feedback
- Refer to the marking rubric to understand what will be evaluated during the final presentation, under both the system demonstration and the final report sections.

Final Report Instructions

- **File Type**: PDF
- **Length**: Maximum 25 pages
- **Submission Mode**: One report per group. Submit soft copy through the course platform

Final Report Structure

Section	Details
1. Cover Page *(1 page)*	Include the following details on the first page of your report: • Project Title • Group Members' Names and Student IDs • Module Name and Code (e.g., SE2030 – Software Engineering) • Submission Date
2. Introduction *(~1-2 pages)*	Provide a clear overview of the project. Include the following: • **Project Overview**: Briefly describe what your project is about. • **Objectives**: State the main goals you aim to achieve through the system. • **Target Users or Stakeholders**: Identify who will use the system or who is affected by it. • **Scope and Limitations**: Clearly define what the system will include and what is out of scope.
3. Requirements *(~2-3 pages)*	Present the gathered requirements for the system in a structured format. Include: • **Functional Requirements**: What the system must do (e.g., user login, search functionality). • **Non-functional Requirements**: Quality aspects such as performance, usability, and security.

	• **Constraints or Assumptions**: Any limitations or assumptions made (e.g., must run on Android, assumes users have internet access).
4. Design *(~4 - 6 pages)*	Describe how the system is planned and structured before coding begins. Include: • **Use Case Diagram**: A visual representation of user interactions with the system. • **Database Design / ER Diagram**: Show how data is structured and related. • **UI Sketches/Screenshots and Descriptions**: Provide screenshots, wireframes, or written descriptions of the user interface.
5. Implementation *(~2 - 3 pages)*	Explain how the system was built. Include: • **Tools and Technologies Used**: Mention programming languages, IDEs, libraries, frameworks, etc. • **Key Features Developed**: List core features that were implemented. • **Screenshots of Core Functions**: Add images showing working parts of your system.
6. Project Management *(~1–2 pages)*	Describe how the team organized the development process. Include: • **Agile Approach and Sprint Summary**: Show how sprints were planned and executed. • **Task Distribution Among Team Members**: Explain who did what. • **Development Phases or Timeline**: Provide a summary of key milestones and their timelines.
7. Conclusion & Future Work *(~1- 2 pages)*	Summarize the project outcome and reflect on future improvements. Include: • **Summary of Achievements**: What was completed successfully. • **Challenges Faced**: Key difficulties or blockers during the project.

	• **Suggestions for Improvement or Extension**: What could be done better or added if more time/resources were available.
8. Individual Contribution, Teamwork & Lessons Learned *(~1- 2 pages)*	In this section, each student should reflect on their **personal involvement** and **team experience** during the project. Please include the following: • **Your specific role** in the project: What tasks or responsibilities did you handle? • **Challenges you faced**: Describe any difficulties you encountered during the project. • **How you overcame those challenges**: Explain the steps you took to solve the problems. • **Key lessons learned**: What did you learn from working on this project? (about teamwork, tools, time management, etc.) • **Reflection on the project**: What aspects of the project went well? What could have been improved? **Note:** • Use a table where each row represents a group member's reflection. • Each student in the group must complete one row, explaining their own role and experience in the project.
9. References *(~1- 2 pages)*	List all the external resources you referred to in your project. Include: • Books, websites, tools, APIs, and libraries. • Follow the **IEEE referencing style** strictly.
10. Appendix *(optional) – Max 2 pages*	Include any supplementary materials. You may attach: • GitHub or repository link to your source code • Additional diagrams • Meeting minutes, sprint logs, or planning documents

Marking Rubrics

Project Proposal Presentation and Report - Marking Rubric

Total: 100 Marks (Presentation: 70 | Report: 30)

Proposal Presentation (70 marks)					
Criteria (Total Marks)	**Excellent**	**Good**	**Satisfactory**	**Needs Improvement**	**Poor**
Problem Analysis (8 marks)	Clear, concise explanation of problem, purpose, goals, and background (6–8 marks)	Clear explanation with minor omissions (5–6 marks)	Covers basic context, lacks clarity in purpose (3–5 marks)	Lacks detail and clarity (2–3 marks)	Not understandable or irrelevant (0–2 marks)
System Overview Diagram (8 marks)	Well-structured, easy-to-understand diagram showing components and relationships (6–8 marks)	Mostly clear diagram with minor issues (5–6 marks)	Diagram exists but lacks clarity or completeness (3–5 marks)	Poorly presented diagram (2–3 marks)	No meaningful diagram (0–2 marks)
Functional Requirements (10 marks)	All major functions well defined, relevant, and clearly explained (8–10 marks)	Most functions defined with minor gaps (6–8 marks)	Basic functions mentioned but not clearly elaborated (4–6 marks)	Few unclear or irrelevant functions (2–4 marks)	Requirements missing or incorrect (0–2 marks)
Non-Functional Requirements (7 marks)	Quality attributes like usability, performance, and security	Mostly relevant quality aspects included (4–6 marks)	Some quality attributes covered (3–4 marks)	Limited or unclear attributes (1–3 marks)	Missing or incorrect content (0–1 marks)

	clearly presented (6–7 marks)				
Stakeholder Identification (7 marks)	All relevant stakeholders identified and roles explained (6–7 marks)	Most stakeholders identified (4–6 marks)	Basic stakeholder list with limited role explanation (3–4 marks)	Vague or incomplete stakeholder info (1–3 marks)	No stakeholder information (0–1 marks)
Assigned Functions (6 Major + Minor) (10 marks)	6 major functions mapped to users and tasks + minor functions explained (8–10 marks)	Functions assigned with minor gaps (6–8 marks)	Most functions assigned but roles not clearly defined (4–6 marks)	Few functions assigned, unclear ownership (2–4 marks)	No functional mapping (0–2 marks)
System Limitations / Constraints (7 marks)	Realistic, clearly explained limitations and constraints (6–7 marks)	Reasonable limitations included (4–6 marks)	Basic limitations mentioned but not discussed (3–4 marks)	Vague or incomplete constraints (1–3 marks)	No mention of limitations (0–1 marks)
Team Coordination & Delivery (7 marks)	Smooth transitions, equal participation, well-managed time (6–7 marks)	Minor issues in time or transitions (4–6 marks)	Unequal participation or unbalanced timing (3–4 marks)	Poor coordination, evident confusion (1–3 marks)	No coordination or disorganized (0–1 marks)
Communication Skills (6 marks)	Clear, confident speaking; professional language and effective visuals (5–6 marks)	Mostly clear with minor errors (4–5 marks)	Adequate presentation; some language/clarity issues (2–4 marks)	Unclear or unconfident delivery (1–2 marks)	Difficult to understand (0–1 marks)
Report (30 marks)					

Criteria (Total Marks)	Excellent	Good	Satisfactory	Needs Improvement	Poor
Introduction (4 marks)	Clear and detailed project overview. Objectives, users, scope, and limitations are well-defined and easy to understand. (4 marks)	Mostly clear with minor missing details. (3–3.5 marks)	Basic overview, lacks depth in objectives or scope. (2–2.5 marks)	Weak explanation, vague or unclear in several parts. (1.5–2 marks)	Very minimal, unclear, or irrelevant introduction. (1 mark)
System Overview Diagram (3 marks)	Clear and relevant diagram showing high-level system workflow, visually easy to follow. (3 marks)	Mostly clear diagram, minor missing labels or details. (2.5 marks)	Basic diagram present but lacks clarity or detail. (2 marks)	Poorly structured or incomplete diagram. (1.5 marks)	Very minimal, unclear, or irrelevant diagram. (1 mark)
Functional Requirements (3 marks)	Well-structured, complete list of main functions with clear descriptions of purpose. (3 marks)	Mostly clear, some functions lack detail. (2.5 marks)	Basic list, unclear or incomplete functions. (2 marks)	Limited or vague listing, lacks explanation. (1.5 marks)	Very minimal or irrelevant functions listed. (1 mark)
Non-Functional Requirements (3 marks)	Clearly defined quality aspects (performance, usability,	Mostly clear, minor missing attributes. (2.5 marks)	Basic non-functional points listed, lacks clarity. (2 marks)	Limited or vague non-functional requirements. (1.5 marks)	Very minimal or unclear statements. (1 mark)

	security, etc.) relevant to the system. (3 marks)				
Stakeholders / Users (3 marks)	All major user roles identified and described with their expected interactions with the system. (3 marks)	Mostly clear, minor missing roles. (2.5 marks)	Basic user roles listed with little description. (2 marks)	Limited or unclear list of stakeholders. (1.5 marks)	Minimal or irrelevant information given. (1 mark)
Major Functions (5 marks)	Six major functions described in detail: who uses it, what it does, expected outcome clearly stated for each. (5 marks)	Mostly complete with small gaps in some functions. (4–4.5 marks)	Basic descriptions provided, several missing details. (3–3.5 marks)	Limited or vague functions with poor explanation. (2–2.5 marks)	Very minimal or unclear descriptions. (1 mark)
System Limitations / Constraints (3 marks)	Clear and realistic constraints or assumptions identified, relevant to the system design. (3 marks)	Mostly clear, minor missing details. (2.5 marks)	Basic points mentioned, not fully explained. (2 marks)	Limited or vague constraints listed. (1.5 marks)	Very minimal or unclear statements. (1 mark)
Timeline & Conclusion (3 marks)	Well-structured timeline table with weekly	Mostly clear, minor missing details in	Basic timeline and conclusion, lacks depth	Limited or vague timeline and unclear	Very minimal or poorly written timeline and

	milestones, clear summary highlighting importance of project and team's completion approach. (3 marks)	timeline or conclusion. (2.5 marks)	or clarity. (2 marks)	conclusion. (1.5 marks)	conclusion. (1 mark)
Written Communication (3 marks)	Well-written report with logical flow, no grammar errors, and proper formatting throughout. (3 marks)	Mostly well-written, minor grammar or formatting issues. (2.5 marks)	Mostly well-written, minor grammar or formatting issues. (2.5 marks)	Weak writing skills, frequent errors, unclear sentences. (1.5 marks)	Very minimal writing quality, **hard to read or poorly structured. (1 mark)**

Progress Presentation - Marking Rubric

Total: 100 Marks (System Demonstration: 50 | Design Document: 50)

System Demonstration (50 marks)					
Criteria	**Excellent**	**Good**	**Satisfactory**	**Needs Improvement**	**Poor**
UI Design (10 marks)	Most planned screens are completed with a clean, consistent, and professional design. Navigation is smooth, visually appealing, and user-friendly. (9–10 marks)	UI mostly complete and easy to navigate, minor layout or visual issues present. (7–8 marks)	Basic screens available but incomplete or inconsistent, reducing usability. (5–6 marks)	UI partially developed with poor layout, unclear navigation, or missing screens. (3–4 marks)	Very minimal UI progress, most screens missing or not functional enough to navigate. (1–2 marks)
CRUD Operations (15 marks)	CRUD features (Create, Read, Update, Delete) working correctly for most modules, smooth data handling	Most CRUD operations functional, minor bugs or incomplete features. (10–12 marks)	Basic CRUD operations work only for some modules, inconsistent results. (7–9 marks)	CRUD operations partially working, frequent errors, missing main features. (4–6 marks)	Very minimal CRUD functionality, most operations fail or are not implemented. (1–3 marks)

	with no major errors. (13–15 marks)				
Database Connection (10 marks)	Stable DB connection established, data stored and retrieved correctly in all tested cases. (9–10 marks)	Connection works with rare issues or occasional delays. (7–8 marks)	Basic connection functional but sometimes fails or incomplete queries. (5–6 marks)	Unstable connection, frequent errors in storing/retrieving data. (3–4 marks)	Very weak or non-functional connection, store/retrieve mostly fails. (1–2 marks)
Validations (5 marks)	All important input fields validated, preventing empty or incorrect data entries, improving data quality. (5 marks)	Most key fields validated, minor gaps in coverage. (4 marks)	Basic validations implemented, some inputs unchecked, allowing errors. (3 marks)	Few validations applied, most inputs not validated, frequent invalid data. (2 marks)	Very minimal or no validations visible during demo. (1 - 0 marks)
Teamwork (5 marks)	All members work well together, share equal responsibility, and support each other	Good teamwork, minor gaps in contribution balance or transitions. (4 marks)	Basic teamwork, some members dominate, uneven involvement. (3 marks)	Limited collaboration, roles unclear, poor coordination in demo. (2 marks)	Very minimal teamwork, one or two members present almost everything, others

	during the demo. Smooth handovers between presenters. (5 marks)				inactive. (1 mark)
Oral Communication Skills (5 marks)	All members speak clearly, confidently, and professionally, use correct technical terms, and answer questions directly. (5 marks)	Mostly clear communication, minor hesitation or wording issues. (4 marks)	Understandable but lacking detail or confidence, occasional unclear points. (3 marks)	Often unclear, hesitant, or struggle to explain features well. (2 marks)	Explanations very minimal, unclear, or incorrect, hard for audience to follow. (1 mark)

Design Document (50 marks)					
Criteria	**Excellent**	**Good**	**Satisfactory**	**Needs Improvement**	**Poor**
Agile Sprint Summary (10 marks)	Detailed summary of sprints with clear task allocation, completed tasks, backlog tracking, and progress	Mostly clear, minor omissions in task tracking or backlog details. (7–8)	Basic sprint summary, lacks detailed task allocation or progress explanation. (5–6)	Limited or vague sprint documentation, unclear task distribution. (3–4)	Very minimal or unclear sprint summary provided. (1–2)

	clearly documented. (9–10)				
Use Case Diagram (10 marks)	Accurate and complete diagram with all actors, use cases, relationships, and proper labeling. (9–10)	Mostly correct with minor errors or missing use cases. (7–8)	Basic diagram with some missing elements or unclear relationships. (5–6)	Poorly structured, several inaccuracies, confusing relationships. (3–4)	Very minimal or irrelevant diagram provided. (1–2)
Class Diagram (10 marks)	Well-structured diagram showing key classes, attributes, methods, and relationships clearly and accurately. (9–10)	Mostly correct diagram with few relationship or naming errors. (7–8)	Basic class diagram present, lacks details or some errors. (5–6)	Poorly designed, unclear or incomplete relationships. (3–4)	Very minimal or unclear diagram. (1–2)
Activity Diagrams (10 marks)	Clear and logical flows for each major function, with well-labeled steps and	Mostly correct diagrams with minor missing flows or	Basic activity diagrams, lacking clarity or depth in	Limited or unclear diagrams with incomplete flows. (3–4)	Very minimal or irrelevant activity diagrams. (1–2)

	decisions, showing correct process logic. (9–10)	details. (7–8)	several areas. (5–6)		
Ethical Considerations (10 marks)	Detailed discussion on data privacy, accessibility, fairness, security, with solutions or mitigation approaches clearly proposed. (9–10)	Most important ethical issues identified, some lacking depth. (7–8)	Basic mention of ethical points, lacks explanation or solutions. (5–6)	Few unclear ethical points, vague or generic discussion. (3–4)	Very minimal or no meaningful ethical considerations included. (1–2)

Final Presentation and Final Report - Marking Rubric

Total: 100 Marks (System Demonstration and Viva: 70 | Final Report: 30)

System Demonstration and Viva (70 marks)					
Criteria	**Excellent**	**Good**	**Satisfactory**	**Needs Improvement**	**Poor**
User Interface Design & Usability(10 marks)	Clean and professional design. All screens complete with consistent layout, colors, and navigation flow. User-friendly and	Well-designed screens with good layout and navigation. Minor inconsistencies or small visual issues but overall	Basic screen design present with some alignment or flow issues. Limited visual appeal and	Incomplete screens with poor layout and unclear navigation. Design looks unpolished and not user-	Very minimal or unclear UI. Screens missing, broken navigation, or poor visual presentation.(0–2)

	visually appealing interface that meets requirements perfectly. (8–10)	user-friendly.(6–8)	usability.(4–6)	friendly.(2–4)	
CRUD Operations & Input Validation(12 marks)	All CRUD operations (create, read, update, delete) work flawlessly across all modules. Data is stored and retrieved accurately in real time. Strong input validations implemented on all relevant fields, preventing invalid or harmful data entries.(10–12)	Most CRUD operations functional with only minor errors. Input validations applied to most fields with few gaps or inconsistencies. Overall reliable functionality.(8–10)	Basic CRUD operations working for some modules, but several incomplete or partially working. Limited validation checks; some invalid data accepted.(6–8)	CRUD functions often fail or are unreliable. Minimal validation on forms, allowing frequent incorrect data entry. User experience disrupted by errors.(3–6)	Very minimal CRUD implementation, most operations fail or are missing. Little to no validation, leading to frequent data errors or broken application flow.(0–3)
DB Connectivity (5 marks)	Connects reliably to the database; store and retrieve work smoothly through the UI with no	Stable connection; store/retrieve work with only minor, infrequent issues.(4)	Connection works but inconsistent; occasional failures or retries needed for store/retrieve.(3)	Unstable connection; frequent errors when trying to store or retrieve data via the UI.(2)	Connection is very weak; store/retrieve mostly fail or require manual workarounds (but some minimal connectivity

	errors during the demo.(5)				is shown) OR No DB connection (0–1)
Integration & Stability (10 marks)	All modules/screens are seamlessly integrated. Navigation is smooth, data flows correctly between features. The system runs without errors or crashes during the entire demo.(8–10)	Modules mostly integrated with minor linking or data flow issues. The system is stable with only occasional, small errors.(6–8)	Some modules are partially connected, causing inconsistent navigation or data flow. A few noticeable errors occur during the demo.(4–6)	Many modules are not well-integrated, broken links or repeated crashes disrupt workflow. System stability is low. (2–4)	Very minimal integration; modules appear isolated with frequent crashes or severe errors making the system hard to use. OR No integration (0–2)
Application of Minimum 1 Design Patterns (5 marks)	At least 2 relevant patterns are implemented correctly, well-integrated across modules, and clearly justified with code references (where/why used and benefits).(5)	patterns applied and generally correct, but with minor implementation gaps or brief justification. Integration is mostly consistent.(4)	At least 1 pattern is applied correctly; the second is partial/weak or not clearly integrated. Justification is basic.(3)	Patterns are attempted but misapplied or not relevant to the problem. Justification is unclear and integration is poor.(2)	Minimal evidence of pattern use (e.g., mentioned but not implemented, or superficial usage). Justification missing or very weak. OR No patterns used (0 -1)
Teamwork & Demo Delivery (8 marks)	Demo flows smoothly with clear handovers between team	Well-coordinated demo with minor timing or transition	Basic coordination; some uneven participation or	Poor coordination, unclear role distribution. Frequent	One or two members present most of the demo while others are

	members. Everyone knows their role, contributes equally, and delivers confidently within the allocated time. (7–8)	issues. Most members actively contribute and show good preparation .(5–6)	awkward transitions. Presentation runs but lacks smooth flow.(3–4)	pauses, confusion, or repeated attempts during the demo. (1–2)	passive. Flow is very disorganized or incomplete. OR No teamwork evident (0 - 1)
Individual Contribution & Code Understanding(15 marks)	Excellent explanations. Can answer any question directly. Know the code fully. Answer questions confidently (9–10)	Good explanations. Can answer most questions clearly. Understands the code well. Responds with minor hesitation.(7–8)	Basic explanations. Answers some questions but lacks detail. Understands only parts of the code. Shows some uncertainty when responding. (5–6)	Limited explanations. Struggles to answer several questions. Partial understanding of the code. Responses are unclear or hesitant.(3–4)	Very minimal explanations. Cannot answer most questions. Shows little understanding of the code. Responses are incomplete or incorrect. (0–2)
Oral Communication Skills (5 marks)	Speaks clearly, confidently, and professionally. Uses appropriate tone and technical vocabulary. Maintains good pace, engages the audience, and delivers without	Speech is mostly clear with minor hesitation or filler words. Uses technical terms appropriately. Pace and tone are generally good.(4)	Speech is understandable but sometimes unclear, hesitant, or monotone. Limited use of technical vocabulary. (3)	Speech is often unclear, too fast or too slow, with frequent pauses or poor word choice. Hard for the audience to stay engaged.(2)	Speech is very unclear, disorganized, or lacking confidence. Audience finds it difficult to follow or understand the message.(0–1)

	hesitation. (5)				
Final Report (30 marks)					
Criteria	**Excellent**	**Good**	**Satisfactory**	**Needs Improvement**	**Poor**
Introduction (4 marks)	Clear and detailed project overview. Objectives, users, scope, and limitations are well-defined and easy to understand. (4)	Mostly clear with minor omissions. (3–3.5)	Basic overview, lacks depth or misses key elements. (2–2.5)	Weak and unclear introduction with limited relevance. (1.5–2)	Very minimal or unclear introduction. (1)
Requirements (4 marks)	Functional, non-functional, and constraints well-defined and structured. (4)	Mostly clear, few missing elements. (3–3.5)	Basic list of requirements, lacks clarity. (2–2.5)	Limited or vague description of requirements. (1.5–2)	Very minimal or unclear requirements. (1)
Design (6 marks)	Accurate and complete Use Case, ER diagram, and UI sketches, clearly explained. (5–6)	Mostly correct with minor errors. (4–5)	Basic diagrams, missing depth or relationships. (3–4)	Poorly structured or unclear diagrams. (1.5–2)	Very minimal or hard-to-understand design. (1)
Implementation (4 marks)	Clear explanation of tools, features, and screenshots	Mostly described, few missing details. (3–3.5)	Basic information provided only. (2–2.5)	Limited or unclear description of implementation. (1.5–2)	Very minimal or unclear explanation. (1)

	of key functions. (4)				
Project Management (3 marks)	Agile approach, sprint summaries, and task distribution clearly explained with a timeline of milestones. (3)	Mostly clear with minor omissions. (2.5)	Basic timeline or tasks listed. (2)	Limited or unclear planning details. (1.5)	Very minimal or no evidence of project management. (1)
Conclusion & Future Work (3 marks)	Strong summary, key challenges addressed, and realistic suggestions for improvement. (3)	Mostly clear with minor gaps. (2.5)	Basic summary with limited depth. (2)	Limited or vague conclusion. (1.5)	Very minimal or unclear closing statements. (1)
Contribution & Lessons Learned (3 marks)	Meaningful, realistic reflections on contributions, challenges, solutions, and lessons learned. (3)	Reflections mostly complete with minor gaps. (2.5)	Basic reflections, lacks depth or clarity. (2)	Limited or vague contributions and learnings. (1.5)	Very minimal or unclear reflections provided. (1)
Written Communication (3 marks)	Report is well-written, logically structured, free of major grammar errors, and formatted properly. (3)	Mostly well-written with minor grammar or formatting issues. (2.5)	Understandable but frequent grammar issues or poor flow. (2)	Weak writing skills, unclear sentences, many errors. (1.5)	Very minimal writing quality, hard to read. (1)

Submission Format

- All reports must be in PDF format.
- All source code should be submitted via GitHub with access granted.
- One group member will submit the documents on behalf of the team.
- Deadlines will be published on Course Web.

Important Notes

- Late submissions will incur penalties.
- All students must contribute equally.
- Ethical risks must be identified early (e.g., data privacy, AI misuse).
- You may apply the knowledge gained from the OOP module, along with your own self-study, to develop the web-based application.

Special Note: Use of AI Tools

- **AI tools may be used as a learning and support resource throughout this project. Students may use AI tools for requirements gathering, brainstorming ideas, exploring solution alternatives, understanding technologies, improving documentation, generating prototypes, debugging, and learning programming concepts.**
- **However, AI tools must not replace the student's own understanding and contribution. All students are expected to fully understand, explain, and defend their individual work, including requirements, design decisions, implementation, testing, and documentation, during project evaluations, presentations, and viva examinations.**
- **Students must ensure that all submitted work is reviewed, validated, and appropriately modified based on their own understanding. Any work that cannot be adequately explained by the student will be treated as unauthorised assistance. Plagiarism, academic misconduct, or excessive reliance on AI-generated content may result in zero marks and disciplinary action in accordance with university regulations.**

---------- **End of the Assignment Specification**----------